

CONTRACTOR SUBMITTAL TRANSMITTAL FORM REV. A

Owner: County of Hawaii
Contractor: Nan, Inc. **Project No.:** WW-4705R
Project Name: Hilo WWTP Phase 1 **Submittal Number:**
Submittal Title: For Information Only
TO:
From: Nan Inc.

Specification No. and Subject of Submittal / Equipment Supplier	
Spec:	Paragraph:
Authored By:	Date Submitted:

Submittal Certification	
Check Either (A) or (B): ☐ (A) We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings with <u>no exceptions</u> . ☐ (B) We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings <u>except</u> for the deviations listed.	
Certification Statement: By this submittal, I hereby represent that I have determined and verified all field measurements, field construction criteria, materials, dimensions, catalog numbers and similar data, and I have checked and coordinated each item with other applicable approved shop drawings and all Contract requirements.	
General Contractor's Reviewer's Signature: 	
Printed Name and Title:	
In the event, Contractor believes the Submittal response does or will cause a change to the requirements of the Contract, Contractor shall immediately give written notice stating that Contractor considers the response to be a Change Order.	
Firm:	Signature: Date Returned:

PM/CM Office Use
Date Received GC to PM/CM: Date Received PM/CM to Reviewer: Date Received Reviewer to PM/CM: Date Sent PM/CM to GC:

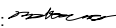
Nan, Inc

PROJECT: HILO WWTP REHABILITATION
AND REPLACEMENT PROJECT - PHASE 1

JOB NO. WW-4705R

THIS SUBMITTAL HAS BEEN CHECKED BY
THIS CONTRACTOR. IT IS CERTIFIED
CORRECT, COMPLETE, AND IN
COMPLIANCE WITH CONTRACT
DRAWINGS AND SPECIFICATIONS. ALL
AFFECTED CONTRACTORS AND
SUPPLIERS ARE AWARE OF, AND WILL
INTEGRATE THIS SUBMITTAL (UPON
APPROVAL) INTO THEIR OWN WORK.

DATE RECEIVED _____
 SPECIFICATION SECTION # _____
 SPECIFICATION _____
 PARAGRAPH _____
 DRAWING _____
 SUBCONTRACTOR _____
 SUPPLIER _____
 MANUFACTURER _____

CERTIFIED BY CQCM or Designee: 

SECTION 15076

PIPE IDENTIFICATION

PART 1 GENERAL

1.01 SUMMARY

- A. Section includes: Pipe identification including the following:
 - 1. Pipe identification markers, flow direction arrows, tags, and bands.
 - 2. Underground warning tape.
 - 3. Tracer wire.
 - 4. Witness markers.
 - 5. Valve Identification.

1.02 REFERENCES

- A. American Society of Mechanical Engineers (ASME):
 - 1. A13.1 - Scheme for the Identification of Piping Systems.

✓1.03 SUBMITTALS

- A. Submit as specified in Section 01330 - Submittal Procedures.
- B. Submit following:
 - ✓1. Product data.
 - 2. Samples.
 - 3. Manufacturer's installation instructions.
 - 4. Submit following as specified in Section 01770 - Closeout Procedures:
 - a. Operation and Maintenance Data.
 - b. Warranty.

PART 2 PRODUCTS

2.01 EXPOSED AND IN-CHASE PIPE IDENTIFICATION

- A. Manufacturers:
 - 1. One of the following:
 - a. Seton, Opti Code Pipe Markers.
 - b. Lab Safety Supply.
 - c. Marking Services, Inc.

B. Materials:

1. Pipe markers: Self-adhesive vinyl, suitable for outdoor application from -40 degrees to 180 degrees Fahrenheit; in accordance with ASME A13.1 requirements:
 - a. Lettering:

Nominal Pipe Diameter	Lettering Size
Less than 1.5	1/2 inch
1.5 inches to 2 inches	3/4 inch
2.5 inches to 6 inches	1-1/4 inches
8 inches to 10 inches	2-1/2 inches
Over 10 inches	3-1/2 inches

- b. Marker colors:

Service	Lettering	Background
Flammables, chemicals, toxics	Black	Yellow
Water, nontoxic solutions, or low hazard liquids	White	Green
Nonflammable or nontoxic gases	White	Blue
Fire quenching fluids (foam, fire water, CO ₂ Halon)	White	Red

2. Coating: As specified in Section 09960 - High-Performance Coatings.
3. Pipe identification tags: Stainless steel with stamped-in 1/4 inch high identifying lettering.
4. Pipe identification tag chains: Stainless steel.
5. Snap-on markers: Markers with 3/4 inch high letters for 3/4 to 4 inch pipe or covering, or 5 inch high letters for 5 inch or larger pipe or cover, as manufactured by one of following:
 - a. Brady, BradySnap-On B-915.
 - b. Seton Setmark.

2.02 BURIED PIPELINE IDENTIFICATION

- ✓ A. Underground warning tape:
1. Manufacturer: One of the following or equal:
 - ✓ a. Seton (of Brady Corporation).
 - b. T. Christy Enterprises, Inc.
 2. Material:
 - a. Polyethylene tape for prolonged underground use.
 - b. Minimum tape thickness: 4 mils.
 - c. Overall tape width: 4 inches.
 - d. Message: "CAUTION" with the name of the service followed by "LINE BURIED BELOW." in black lettering on colored background in accordance with approved APWA colors.
 - 1) Water: Blue.

- 2) Sewer: Green.
- 3) Telephone: Orange.
- 4) Gas and other services: Yellow.

B. Tracer wire:

- 1. Manufacturers: One of the following or equal:
 - a. Kris-Tech Wire.
 - b. Aegion Corrpro.
- 2. Materials: One of the following or equal:
 - a. Solid copper conductor
 - b. Thickness minimum: 10 gauge.
 - c. Insulation:
 - 1) Match insulation color to the color of the pipe being installed.
 - 2) UF type, direct bury.
 - 3) 30 mil HMWPE.
- 3. Splicing kit:
 - a. Manufacturers: One of the following or equal:
 - 1) 3M, Model 82 A1N.
- 4. Station box:
 - a. Lid and collar materials: Cast iron.
 - b. Lid type: Locking.
 - c. Able to withstand heavy traffic loading.
 - d. Manufacturers: One of the following or equal:
 - 1) CP Test & Valve Products, Inc., Glenn Test Station.
 - 2) Farwest Corrosion Control.

C. Witness markers:

- 1. Manufacturers: One of the following or equal:
 - a. Carsonite Composites, Utility Marker.
 - b. Hampton Technical Associates, Inc.
- 2. Materials:
 - a. Glass fiber and resin reinforced thermosetting composite material.
 - b. UV resistant.
- 3. Constructed as a single piece.
- 4. Pointed at the bottom end.
- 5. Information to be included on the marker:
 - a. "Caution" (type of service) "Pipeline".
 - b. Phone number for Underground Service Alert.
 - c. Phone number for the Engineer in case of emergency.
 - d. Station number.
 - e. Offset:
 - 1) Only provide offset if marker is not directly over the pipe.
 - f. Name of appurtenance or fitting (e.g., 45, BO, ARV, etc.).

2.03 VALVE IDENTIFICATION

A. The Contractor shall furnish and install tags for all valves and gates required for the Work:

- 1. Tags shall be 2-inch diameter round, stainless steel or brass for buried applications.

2. Tags shall be furnished with a non-corrosive metal wire suitable for attaching the tag to the operator base.
3. Tags shall be stamped in 1/4-inch high letter:
 - a. Tags shall not be attached in such a way as to inhibit the operation of the valve or gate.
4. Buried valve tags shall be secured to concrete s with the specified valve or gate number.
5. Submit 2 samples of the type of tag proposed and the manufacturer's standard color chart and letter styles to the Engineer for review.
6. Manufacturer: The following or approved equal:
 - a. Seton Name Plate Company, Branford, CT.

PART 3 EXECUTION

3.01 EXAMINATION

- A. Verify satisfactory conditions of substrate for applying identification.
- B. Verify that conditions are satisfactory for installation and application of products as specified in Section 01600 - Product Requirements.

3.02 PREPARATION

- A. Prepare and coat surfaces as specified in Section 09960 - High-Performance Coatings.
- B. Prepare surface in accordance with identification product manufacturer's instructions.

3.03 EXPOSED AND IN-CHASE PIPING IDENTIFICATION

- A. Identify exposed piping, valves, and accessories, and piping, valves, and accessories in accessible chases with lettering or tags designating service of each piping system with flow directional arrows and color code.
- B. Color code:
 1. Paint all piping with colors as scheduled in Piping Color Code and Marker Schedule.
- C. Lettering and flow direction arrows:
 1. Stencil lettering on painted bands or use snap-on markers on pipe to identify pipe. When stenciling, stencil 3/4-inch high letters on 3/4- through 4-inch pipe or coverings, or 5-inch high letters on 5-inch and larger pipe or coverings.
 2. Provide lettering and flow direction arrows near equipment served, adjacent to valves, both sides of walls and floors where pipe passes through, at each branch or tee, and at intervals of not more than 50 feet in straight runs of pipe.
- D. Where scheduled, space 6-inch wide bands along stainless steel pipe at 10-foot intervals and other pipe at 5-foot intervals.

- E. Label chemical tank fill pipelines at locations which are visible from chemical fill stations.
- F. Metal tags:
 - 1. Where outside diameter of pipe or pipe covering is 5/8 inch or smaller, provide metal pipe identification tags instead of lettering.
 - 2. Fasten pipe identification tags to pipe with chain.
 - 3. Where tags are used, color code pipe as scheduled.

3.04 BURIED PIPING IDENTIFICATION

- A. Underground warning tape:
 - 1. Non-detectable warning tape:
 - a. Place continuous run of warning tape in pipe trench, 12 inches above the pipe.
 - ✓ 2. Detectable warning tape:
 - a. Place continuous run of warning tape in pipe trench, 12 inches above the pipe or a greater height if necessary to limit the tape bury depth to 36 inches. Do not bury detectable warning tape deeper than 36 inches.
- B. Tracer wire:
 - 1. Install on all non-metallic pipe.
 - 2. Install an electrically continuous run of tracer wire along the entire length of the pipe with wire terminations in valve boxes, vaults, or structures.
 - 3. Install tracer wire on top of the pipe and secure to pipe with tape a minimum of every 10 feet.
 - 4. Where approved by the Engineer, splice sections of wire together using approved direct bury wire nuts.
 - a. Twisting the wires together is not acceptable.
- C. Witness markers:
 - 1. Install over pipe in unpaved open-space areas at intervals not greater than 200 feet.
 - 2. Place markers at appurtenances located in unpaved areas.
 - 3. Embed markers at least 18 inches into the soil.

3.05 APPLICATION

- A. Identify piping with legend markers, directional arrow markers, and number markers; use self-adhesive arrow roll tape to secure ends of piping markers and indicate flow direction.
- B. Provide legend markers, directional arrow markers, and number markers where piping passes through walls or floors, at piping intersections and at maximum 15 foot spacing on piping runs.
- C. Provide piping marker letters and colors as scheduled.
- D. Place markers on piping so they are visible from operator's position in walkway or working platform near piping. Locate markers along horizontal centerline of pipe, unless better visibility is achieved elsewhere.

3.06 PIPING COLOR CODE AND MARKER SCHEDULE

Pipe Service Designation (Piping Schedule)	Pipe Color	Color of Letters	Marker Legend
1W	ANSI safety blue (5150)	White	Potable Water (1W)
2W	Light blue with black bands	White	Non-Potable Water (2W)
3W	ANSI safety purple (5575)	White	Chlorinated Final Effluent (3W)
BP	Black (C900)	White	Building Plumbing (BP)
BSL	ANSI safety brown (2284) with black bands	White	Blended Sludge (BSL)
CD	Black (C900) with blue bands	White	Chemical Drain (CD)
CEN	Light Gray (C705) with white bands	Black	Centrate (CEN)
CLS	ANSI Safety Yellow (5626)	Black	Chlorine Solution (CLS)
CSL	ANSI safety brown (2284) with black bands	White	Cold Sludge (CSL)
D	Black (C900)	White	Drain (D)
DG	ANSI safety yellow (5625) with red bands	Black	Digester Gas (DG)
DSL	ANSI safety brown (2284) with black bands	White	Digested Sludge (DSL)
FD	Black (C900)	White	Floor Drain (FD)
FIL	Light green	Black	Filtrate (FIL)
FM	Light gray (C705)	Black	Force Main (FM)
FW	ANSI safety red (5525)	White	Fire Water (FW)
GRR	ANSI safety white (1864)	Black	Glycol Reservoir Return (GRR)
GRS	ANSI safety white (1864)	Black	Glycol Reservoir Supply (GRS)
GRT	Silver	Black	Grit Slurry (GS)

Pipe Service Designation (Piping Schedule)	Pipe Color	Color of Letters	Marker Legend
HFR, HFS	ANSI safety white (1864) with red bands	Black	Hydraulic Fluid Return (HFR), Hydraulic Fluid Supply (HFS)
HPA	ANSI safety green (5575)	White	High Pressure Air (HPA)
HPW	Light blue with yellow bands	Black	High Pressure Water (HPW)
HSL	ANSI safety brown (2284) with red bands	White	Heated Sludge (HSL)
HW, HWR, HWS	Light blue with red bands	Black	Hot Water (HW), Hot Water Return (HWR), Hot Water Supply (HWS)
IA	ANSI safety green (5575)	White	Instrument Air (IA)
LPG	ANSI safety red (5525)	White	Propane (LPG)
MSL	ANSI safety brown (2284) with yellow bands	Black	Mixing Sludge (MSL)
NS	Orange	Black	Nutrient Solution (NS)
OF	Light gray with orange bands	Black	Overflow (OF)
PD	Black (C900)	White	Process Drain (PD)
PI	Light gray (C705)	Black	Primary Influent (PI)
POLY	ANSI safety purple (5575) with yellow bands	Black on yellow pipe markers (specify "POLYMER") w/arrows	Neat Polymer (POLY)
POS	ANSI safety purple (5575) with light blue bands	Black on yellow pipe markers (specify "POLYMER") w/arrows	Polymer Solution (POS)
PR	Dark gray	White	Plant Recycle (PR)
PS	ANSI safety brown (2284) with black bands	White	Primary Sludge (PS)

Pipe Service Designation (Piping Schedule)	Pipe Color	Color of Letters	Marker Legend
PSC	ANSI safety brown (2284) with white bands	Black	Primary Scum (PSC)
RD	Black (C900)	White	Roof Drain (RD)
REC	White with green bands	White	Recirculation (REC)
RSS	ANSI safety brown (2284) with orange bands	White	Return Secondary Sludge (RSS)
SAM	Gray	White	Sample (SAM)
SEP	Dark gray with purple bands	White	Septage (SEP)
SPD	Black (C900)	White	Sump Pump Discharge (SPD)
SSC	ANSI safety brown (2284) with light green bands	Black	Secondary Scum (SSC)
SW	Light blue	Black on yellow pipe markers (specify "SEAL WATER") w/arrows	Seal Water (SW)
TD	Black (C900)	White	Tank Drain (TD)
TSL	ANSI safety brown (2284) with yellow bands	Black	Thickened Sludge (TSL)
V	Black	White	Vent (V)
WSS	ANSI safety brown (2284) with orange bands	White	Waste Secondary Solids (WSS)

END OF SECTION



NOTE: Not all widths are available as standard for the stock color/legend combinations shown. Contact Christy's™ for stock availability of specific products. Non-stock combinations are subject to minimum requirements and plate charges.

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Christy's Detectable Marking Tape

Our DETECTABLE marking tape provides for easy buried pipeline detection and below ground identification and warning. The tape can be located below ground with a non-ferrous metal detector when buried at the proper depths. All Christy's tape meets or exceeds the industry standards including the American Public Works Association (APWA) color code.

TAPE MUST BE BURIED FLAT FOR MAXIMUM DETECTABILITY AND LINE PROTECTION. CONSULT YOUR SPECIFIC LOCATOR MANUFACTURER FOR BURIAL LIMITATIONS.

Tape Width	Tape Bury Depth
2"	6" - 9"
3"	6" - 12"
6"	6" - 18"
12" or wider	6" - 24"

SPECIFICATIONS

- Christy's™ DETECTABLE marking tape consists of a 4.5 mil (0.005") thickness; five-ply composition; ultra-high molecular weight; 100 percent virgin polyethylene; acid,alkaline and corrosion resistant.
- The tape tensile strength is in accordance with ASTM D882-80A and will not be less than 7800 PSI.
- Elongation properties are in accordance with ASTM D882-80A and are less than 150% at break point.
- Tape color & legend combination will be in accordance with APWA or local requirements. The color is _____. The legend reads _____.
- **Tape is 5 mil overall thickness** with a .35 mil solid aluminum foil core. A .8 mil clear film, reverse printed with a repeating warning message is laminated to aluminum foil with a 3.75 mil clear film backing.
- Tape width shall be _____. (For best results we recommend a tape width at least ¾ of the diameter of the pipeline being protected. A 4" diameter line would use 3" or wider tape for optimum protection). Standard widths are 2", 3", 6", & 12".
- Tape lettering shall be 1" tall on 2" detectable tape and 1.5" tall on 3"- 12" tape.

SPECIAL AND CUSTOM LEGENDS

Christy's™ offers a full range of non-standard tapes, including specialty applications such as Telemetry, Irrigation and additional water line wording. All standard combinations listed are available in at least one color/width combination. We offer specific agency legends, designations or color combinations. Custom legends can include the use of specific wording, insignias and phone numbers. Call for special pricing.